



**Al-Kunooze University**

**Department of Cyber Security Science**

Report of: Computer Programming I

(CyS101) Class

First Semester (First Stage)

**Title: Internet speed test program code**

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## Abstract

In today's digital age, a reliable internet connection is crucial for both personal and professional activities. Whether streaming videos, participating in online meetings, or simply browsing the web, knowing the speed of your internet connection can significantly impact your experience.

This is where the internet speed test code comes into play. Utilizing the **speedtest** library, this code allows users to easily measure their download and upload speeds, providing valuable insights into their network performance. By executing this simple yet effective tool, users can identify potential issues with their connection, assess the quality of their service provider, and make informed decisions about their internet usage.

In essence, this code serves as an essential resource for anyone looking to understand their internet connectivity better and ensure they are getting the performance they need.

## Table of contents

|                                                |          |
|------------------------------------------------|----------|
| Introduction .....                             | Page (2) |
| Methodology (Python Code) .....                | Page(3)  |
| Results (Running of Python Code) .....         | Page(6)  |
| Discussion of Result of Python Code .....      | Page(7)  |
| Conclusion/recommendations (future work) ..... | Page(8)  |
| References .....                               | Page(8)  |

# I. Introduction

The code used for measuring internet speed relies on the **speedtest** library, which allows users to accurately assess download and upload speeds.

## Key Features:

- **Download Speed Measurement:** The code calculates the speed of data being downloaded from the internet, helping users understand how fast their connection is.
- **Upload Speed Measurement:** It also measures the speed of data being uploaded, which is important for activities such as file sharing and video uploads.

## Benefits:

1. **Performance Assessment:** It helps users evaluate the performance of their internet connection.
2. **Identifying Issues:** The tool can reveal connection problems, assisting in improving user experience.
3. **Service Provider Selection:** It aids in making informed decisions when choosing an internet service provider based on actual performance.

In summary, this code serves as a valuable tool for understanding and improving internet performance.

## II. Methodology (Python Code)

### Explanation of the Code

This code snippet is a simple Python script that uses the speedtest library to measure internet speeds. Here's a breakdown of each part:

#### 1. Importing the Library:

```
python ▶ Run Copy  
from speedtest import Speedtest
```

This line imports the Speedtest class from the speedtest library, which provides methods to perform speed tests.

#### 2. Creating a Speedtest Object:

```
python ▶ Run Copy  
wifi = Speedtest()
```

Here, an instance of the Speedtest class is created and assigned to the variable wifi. This object will be used to conduct the speed tests.

#### 3. Measuring Download Speed:

```
python ▶ Run Copy  
print("Getting Download speed...")  
download = wifi.download()  
print(f"Download: {download / 1024 / 1024:.2f} mbps")
```

- The first line prints a message indicating that the download speed measurement is starting.
- The download() method is called on the wifi object to measure the download speed, and the result is stored in the download variable.

- The final line converts the download speed from bits per second to megabits per second (by dividing by 1024 twice) and prints the result formatted to two decimal places.

#### 4. Measuring Upload Speed:

```
python ▶ Run 📋 Copy  
  
print("Getting Upload speed...")  
upload = wifi.upload()  
print(f"Upload: {upload / 1024 / 1024:.2f} mbps")
```

- Similar to the download section, this part measures the upload speed.
- It first prints a message, then calls the upload() method on the wifi object to measure the upload speed.
- Finally, it converts the upload speed to megabits per second and prints the result.

#### Python code:

```
wifi.py x ⋮  
  
1 from speedtest import Speedtest Reader Mode ✓  
2 wifi = Speedtest()  
3  
4 print("Getting Download speed...")  
5 download = wifi.download()  
6 print(f"Download: {download / 1024 / 1024:.2f} mbps")  
7  
8 print("Getting Upload speed...")  
9 upload = wifi.upload()  
10 print(f"Upload: {upload / 1024 / 1024:.2f} mbps")  
11 💡
```

## New instructions

There are some new instructions (not in the lectures), in the code snippet I provided, download and upload speeds are measured in bits per second (bps). However, to present these speeds in a more user-friendly format, they are converted from bits to megabits (Mbps) using the following conversion:

### 1. Understanding Units:

- **Bit:** The smallest unit of data in computing. Internet speeds are often expressed in bits per second (bps).
- **Byte:** 1 byte = 8 bits. This is an important conversion factor because file sizes are usually measured in bytes.
- **Kilobit (Kb):** 1 Kb = 1,024 bits.
- **Megabit (Mb):** 1 Mb = 1,024 Kb = 1,024 x 1,024 bits = 1,048,576 bits.

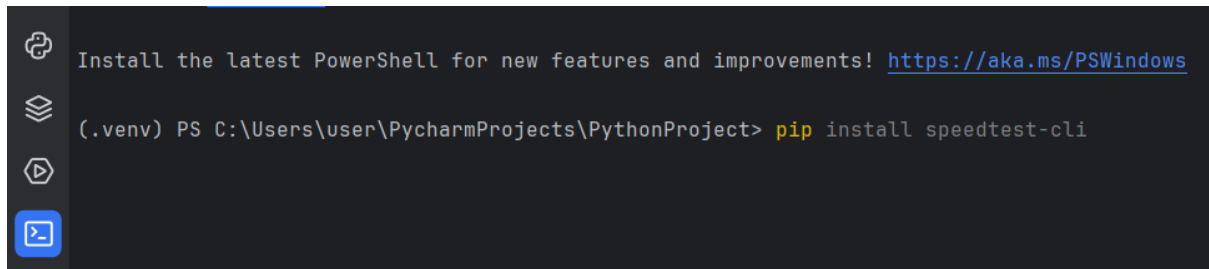
### 2. Conversion in Code:

In the code, the speed measured (in bits) is divided by 1,024 twice to convert it to megabits:

```
python ▶ Run 📋 Copy  
download / 1024 / 1024
```

- The first division by 1,024 converts bits to kilobits:
  - `download / 1024` gives the speed in kilobits (Kb).
- The second division by 1,024 converts kilobits to megabits:
  - `download / 1024 / 1024` gives the speed in megabits (Mb).

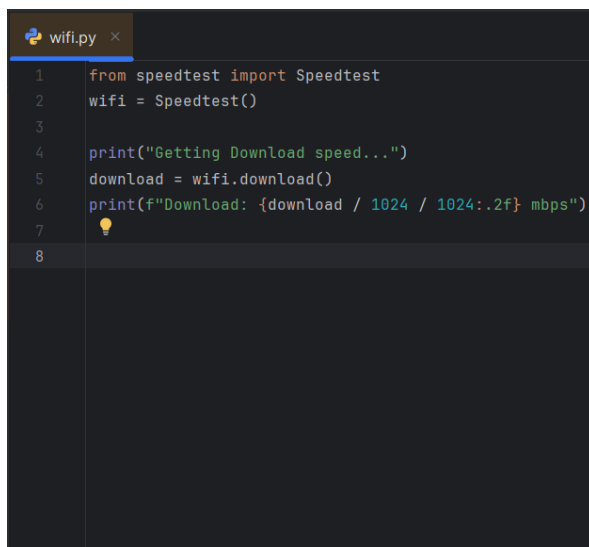
### III. Results (Running of Python Code)



```
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

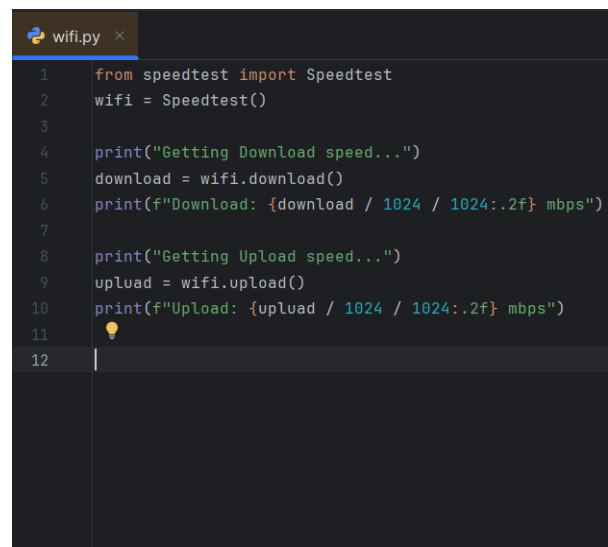
(.venv) PS C:\Users\user\PycharmProjects\PythonProject> pip install speedtest-cli
```

(1)



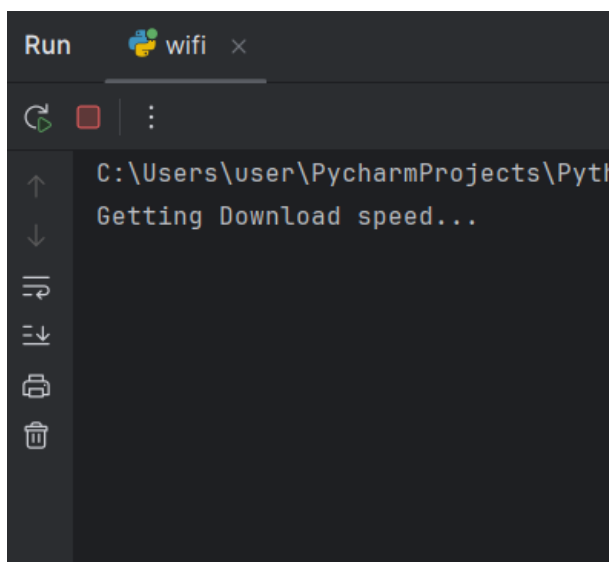
```
wifi.py x
1 from speedtest import Speedtest
2 wifi = Speedtest()
3
4 print("Getting Download speed...")
5 download = wifi.download()
6 print(f"Download: {download / 1024 / 1024:.2f} mbps")
7
8
```

(2)



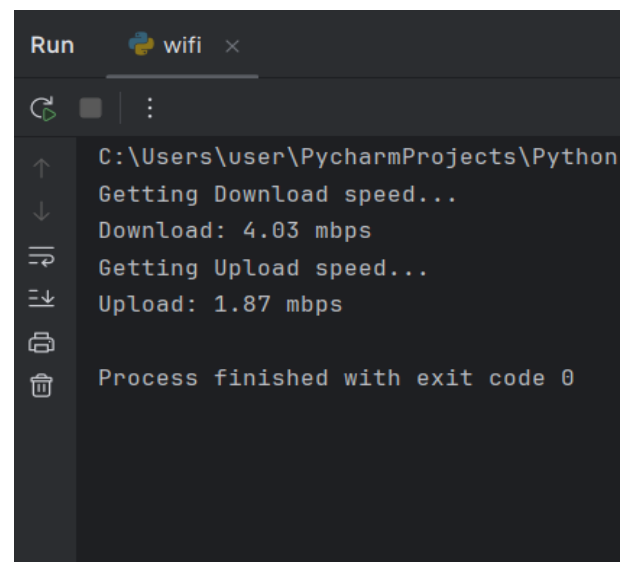
```
wifi.py x
1 from speedtest import Speedtest
2 wifi = Speedtest()
3
4 print("Getting Download speed...")
5 download = wifi.download()
6 print(f"Download: {download / 1024 / 1024:.2f} mbps")
7
8 print("Getting Upload speed...")
9 upload = wifi.upload()
10 print(f"Upload: {upload / 1024 / 1024:.2f} mbps")
11
12
```

(3)



```
Run wifi x
C:\Users\user\PycharmProjects\PythonProject> python wifi.py
Getting Download speed...
```

(4)



```
Run wifi x
C:\Users\user\PycharmProjects\PythonProject> python wifi.py
Getting Download speed...
Download: 4.03 mbps
Getting Upload speed...
Upload: 1.87 mbps
Process finished with exit code 0
```

(5)

## IV. Discussion of Result of Python Code

- 1) In the first image, the speedtest-cli library is installed using the command: `pip install speedtest-cli`
- 2) In the second image, the Speedtest class has been imported from the speedtest library. Which is used to measure speed. Also, a new object called `wifi` has been created from the speedtest class.
- 3) In the third image, the program prints a message stating that it is measuring the download speed.  
The speed is then measured using `wifi.download`, which returns the value in bits per second.  
The value is converted to megabits per second by dividing by 1024 twice (to convert bits to kilobits and then to megabits).  
  
As well as measuring the upload speed, the program prints a message stating that it is measuring the upload speed. The speed is measured using `wifi.upload`, in the same way as in measuring the download speed.
- 4) In the fourth and fifth picture I got the results of measuring the internet speed

### Results:

- Download speed: 4.03 megabits per second (Mbps)
- Upload speed: 1.87 megabits per second (Mbps)

## Conclusion/recommendations (future work)

### Future development of the program

1. Graphical interface: A graphical user interface (GUI) can be added to make the program more user friendly, such as using the Tkinter or PyQt library.
2. Storing results: The ability to save results to a file (such as CSV or JSON) for later analysis or the speed of the Internet reader over time.
3. Advanced settings: Add user options to specify where to measure speed or connection settings.
4. Analysis: Add tools to analyze data, such as weather effects that have affected speed changes over time.
5. Notifications: The ability to send a notification to the user if the speed is below a certain level that is determined for a long time.
6. Integration with other services: Join other network services or analyze them more comprehensively.

 In this way, the program can transform from a simple speed tool into a comprehensive network performance analysis tool.

## References

- 1) Educational resources on building GUIs in Python
- 2) Powered by the latest AI technologies, Poe delivers seamless conversational experiences, enhanced productivity, and creative content generation.
- 3) YouTube